Insight Makers

MANUFACTURING DOWNTIME

TEAM MEMBERS:

- *Nada Alsaeed Alashry (Team Leader)*
- *Fady Mamdouh shokry*
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- *Rana Rabea Shabaan*
- *Manar Ahmed Fouad*
- *Ahmed Abu Esmail*
- *Almostafa Ibrahem*

Objectives

- To understand the primary causes and patterns leading to machine downtime.
- To identify opportunities for improvement and propose actionable corrective measures.

Data Sources

- Two main data files were merged (Downtime records and Machine Information).
- Data Volume: The analysis is based on a dataset containing approximately 10,000 records and 26 columns.

Data Quality and Preparation

- Data Quality: No missing values (Nulls) were found in the critical analysis columns.
- Methodology: The data was cleaned, merged, and prepared for statistical and visual analysis

Manufacturing Downtime Analysis

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Team Workflow and Milestones

1 Brainstorming and Problem Framing

We defined the core problem (downtime) and identified the priority departments and machines for analysis.

2 Defining Research Question

Established a list of questions the data must answer, such as: "What is the most frequent downtime reason?" and "What is the efficiency score of each machine?"

3 Initial Data Cleaning (Excel/Review)

Quick review of the source CSV files in Excel to verify formatting, check delimiters, and address basic inconsistencies before loading the data.

4 Preparation and Structuring (SQL)

Loaded data into the database. Used SQL to ensure data integrity, create table relationships, and perform initial record aggregation

5 Deep Analysis (Python)

Used Pandas and Numpy libraries to perform complex calculations, compute efficiency metrics, and extract key performance indicators (KPIs)

6 Visual Storytelling (Python)

Used Matplotlib and Seaborn to create impactful visual charts that clearly convey the analytical

7 Final Dashboarding (Power BI)

The final results were imported into Power BI to build interactive dashboards, enabling management to easily explore data and make informed decisions.

Our Exact Plan

Methodology and Tool Stack

1. Microsoft Excel 📊

- Objective: Initial Data Review and Source Standardization.
- Impact: Excel was utilized for a quick, initial assessment of the raw CSV files, verifying correct formatting, identifying delimiters, and resolving minor inconsistencies before the data was loaded for heavy-duty processing.

2. SQL (Structured Query Language) 💾

- Objective: Modling and data analysis
- Data Structuring, Integrity, and Initial Aggregation.
- Impact: SQL was crucial for establishing the database schema, enforcing data integrity via foreign key relationships, performing efficient joins between the two data tables, and executing the first round of large-scale aggregations (e.g., calculating the SUM of duration) which formed the analytical backbone.

3. Python (with Pandas/Matplotlib) 🐍

- Objective: Deep Statistical Analysis and Visual Storytelling.
- Impact: Python enabled complex metric calculations (such as the annual Efficiency_Score_year), allowed for advanced comparative analysis across shifts and departments, and was the primary tool for generating high-impact visual plots (Bar Charts, Pie Charts) to communicate findings.

4. Power BI 💻

- Objective: Interactive Dashboards and Executive Reporting.
- Impact: The final, processed results were channeled into Power BI to build dynamic, interactive dashboards. This transformed the static analysis into an easily accessible tool for management to explore the data visually and make prompt, data-driven decisions.

Research Questions

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1 Overall Performance Analysis

- What is the total number of downtime cases over a given period?
- What is the total downtime in each department or machine?
- What is the average downtime duration in the plant?
- What is the percentage of planned vs. unplanned downtime?

2 Machine-level Analysis

- Which machine stops more frequently than others?
- Which machines have the longest downtime durations?
- Is there a relationship between machine type and downtime cause?

3 Department-level Analysis

- Which department (Assembly / Stamping / Painting) has the highest downtime rate?
- What are the common causes of downtime in each department?
- Are there differences between departments in repair time?

4 Root Cause Analysis

- What are the most frequent causes of downtime?
- What is the relationship between cause type and downtime duration?

5 Shift-level Analysis

- Which shift experiences the highest number of downtimes?
- Does downtime duration vary between shifts?
- Are there specific operators with recurring downtimes?

6 Time-based Analysis

- Do downtimes increase on specific days or months?
- Is there monthly downtime pattern?

Excel

We initiated the data analysis process using Microsoft Excel to facilitate preliminary brainstorming and essential data familiarization. This crucial first step involved standardizing variables, integrating key data points, and calculating the core performance metric: Machine Efficiency.

Key Implementation Steps

1 Data Setup (2 Sheets)

- We started with two files: ManufactureDowntime and machine_info

2 Data Understanding & Preparation

- Reviewed both sheets and understood the purpose of each column.
- Checked data quality: no NULL values, no formatting issues, and consistent entries.
- Linked the two sheets using Machine ID.
- Created additional calculated columns based on existing data.

Identified the key questions that will guide the analysis.

3 Data Transformation

- We performed several key steps to prepare the data:
- We converted Down Time from minutes to hours by dividing the original value by 60. This was done to make the time duration easier to understand and use in our yearly analysis.
- We simplified the status labels, replacing the values True and False. True now means Maintenance, and False is Faluier .
- We used XLOOKUP to bring the Target value from machininfo into the main fault sheet. This step was essential to match each machine's actual operation with its set goal

Manufacturing_'Downtime

Column	Description
Downtime_ID	Unique downtime event identifier e.g. DT-00001
Machine_ID	Machine code linking to Machine_Info
Machine_Name	Descriptive machine name
Department	Production department
Start_Time	Timestamp when downtime started (UTC)
End_Time	Timestamp when downtime ended (UTC)
Duration_Minutes	Duration in minutes (End_Time - Start_Time)
Downtime_Reason	Categorized reason for downtime
Operator	Operator on duty
Shift	Shift when downtime occurred (A/B/C)
Planned	Boolean indicating planned maintenance or not
Run time	The amount of time a machine or production line is actively running and producing product (excluding any form of downtime).
Duration_Hour	Duration minut / 60 min
Target_Runtime_Hours_Day	The planned or expected number of hours per day that the equipment or machine should operate under normal production conditions (transfer to manufacturing _downtime by xlookup).
Efficiency score	A measure of how effectively a machine or process produces output compared to its maximum possible performance.

Machine_Info

Column	Description
Machine_ID	Machine code linking to Machine_Info
Manufacturer	The name of the company or organization that produced or supplied the equipment, machine, or component.
Installation year	The calendar year in which the equipment or machine was installed and commissioned for operation.
Target_Runtime_Hours_Day	The planned or expected number of hours per day that the equipment or machine should operate under normal production conditions.
Machine_Age	The number of years the machine has been in operation since installation or commissioning.
Target_Runtime_Hours_year	The planned or expected number of hours the machine should run during the year (pivot tabel)
Total Duration_hours_year	The total number of hours available in the year for the machine, including both running and downtime hours (pivot tabel)
Total Runtime_Hours_year	The actual number of hours the machine operated during the year, excluding downtime (pivot tabel)
Efficiency_Score_year	A metric showing how effectively the machine performed relative to its target runtime or potential output during the year.

SQL's Role II: Initial Aggregation and Querying

Analytical Role of SQL: Beyond structuring, SQL was utilized to perform crucial preliminary aggregations and complex joining operations that guided the direction of the deep analysis in Python.

Value Added: By completing these complex joins and large aggregations in SQL, we significantly reduced the processing load and complexity required in the Python analysis phase.

Analysis

1 Create Database

- Create Database Named Manufacturing Which Contain Tables.

2 Import files

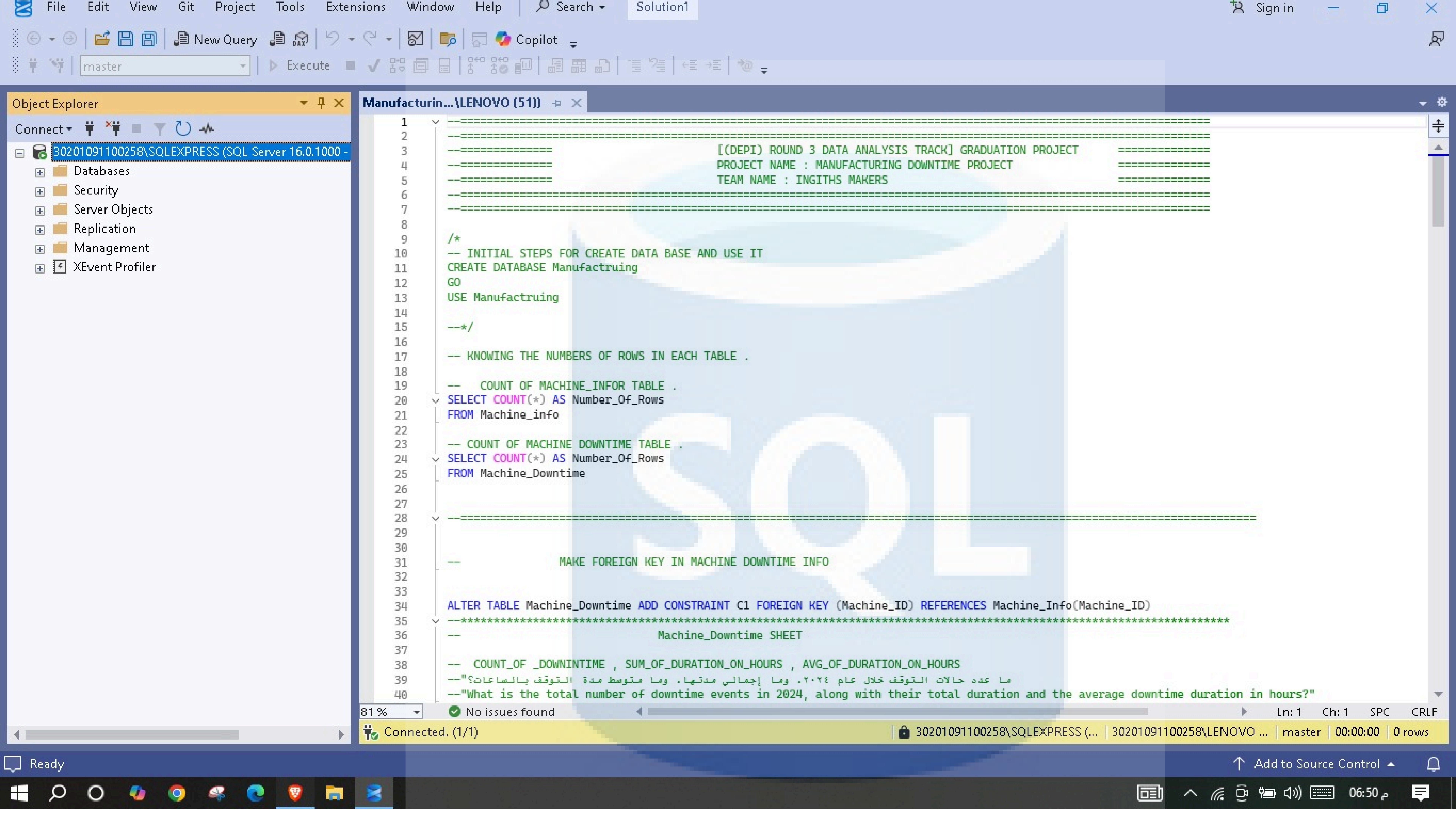
- Import Two CSV Files .
- The First File Is Machine_Downtime .
- The Second File Is Machine_Info .

3 Create Constrint

Creat Constraint Which Connect Tow Table With Each Other By The Column Which Name Is Machin_ID .

4 Make Some Analysis

Write Some Queries To Answer Some Questions .



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Server Objects
Replication
Management
XEvent Profiler
Manufacturin...\LENOVO (51))
--============== [(DEPI) ROUND 3 DATA ANALYSIS TRACK] GRADUATION PROJECT ==============
--============== PROJECT NAME : MANUFACTURING DOWNTIME PROJECT ==============
--============== TEAM NAME : INGITHS MAKERS ==============
/*
-- INITIAL STEPS FOR CREATE DATA BASE AND USE IT
CREATE DATABASE Manufactruing
GO
USE Manufactruing
--*/
-- KNOWING THE NUMBERS OF ROWS IN EACH TABLE .
-- COUNT OF MACHINE_INFOR TABLE .
SELECT COUNT(*) AS Number_Of_Rows
FROM Machine_info
-- COUNT OF MACHINE DOWNTIME TABLE .
SELECT COUNT(*) AS Number_Of_Rows
FROM Machine_Downtime
-- MAKE FOREIGN KEY IN MACHINE DOWNTIME INFO
ALTER TABLE Machine_Downtime ADD CONSTRAINT C1 FOREIGN KEY (Machine_ID) REFERENCES Machine_Info(Machine_ID)
-- Machine_Downtime SHEET
-- COUNT_OF _DOWNINTIME , SUM_OF_DURATION_ON_HOURS , AVG_OF_DURATION_ON_HOURS
--"ما عدد حالات التوقف خلال عام ٢٠٢٤، وما إجمالي مدتها، وما متوسط مدة التوقف بالساعات؟"
--"What is the total number of downtime events in 2024, along with their total duration and the average downtime duration in hours?"
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Object Explorer

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  - Databases
  - Security
  - Server Objects
  - Replication
  - Management
  - XEvent Profiler

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```sql
ALTER TABLE Machine_Downtime ADD CONSTRAINT C1 FOREIGN KEY (Machine_ID) REFERENCES Machine_Info(Machine_ID)
--*******************************************************************************************************************
--                              Machine_Downtime SHEET

--  COUNT_OF _DOWNINTIME , SUM_OF_DURATION_ON_HOURS , AVG_OF_DURATION_ON_HOURS
--"ما عدد حالات التوقف خلال عام ٢٠٢٤. وما إجمالي مدتها. وما متوسط مدة التوقف بالساعات؟"
--"What is the total number of downtime events in 2024, along with their total duration and the average downtime duration in hours?"

SELECT COUNT(Duration_hours) AS COUNT_OF_DOWNTIME, SUM(Duration_hours)  AS SUM_OF_DURATION_ON_HOURS, AVG(Duration_hours) AS AVG_OF_DURATION_ON_HOURS
FROM Machine_Downtime


--=========================================================================================================================



-- COUNT_OF_DOWNTIME , START_DIWB_MONTH , TOTAL_DURATION_HOURS
--"ما عدد حالات التوقف في كل شهر خلال عام ٢٠٢٤. وما إجمالي مدة التوقف بالساعات لكل شهر؟"
--"What is the number of downtime events per month in 2024, along with the total downtime duration in hours for each month?"
SELECT COUNT(Duration_hours) AS COUNT_OF_DOWNTIME, Start_Down_month  ,  SUM(Duration_hours) AS TOTAL_DURATION_HOURS
FROM Machine_Downtime
GROUP BY Start_Down_month
ORDER BY  Start_Down_month


--=========================================================================================================================



--RELATIONSHIP BETWEEN DURATION HOURS AND SHIFT
--"ما عدد حالات التوقف لكل شيفت خلال عام ٢٠٢٤. وما إجمالي مدة التوقف بالساعات لكل شيفت؟"
--"What is the number of downtime events for each shift in 2024, along with the total downtime duration in hours for each shift?"
SELECT COUNT(Duration_hours) AS COUNT_OF_DOWNTIME, Shift ,  SUM(Duration_hours) AS TOTAL_DURATION_HOURS
FROM Machine_Downtime
GROUP BY  SHIFT
ORDER BY TOTAL_DURATION_HOURS DESC


--=========================================================================================================================



-- RELATION BETWEEN DURATION HOURS , SHIFT AND MONTH
```

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```sql
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--RELATIONSHIP BETWEEN DURATION HOURS AND SHIFT
--"ما عدد حالات التوقف لكل شيفت خلال عام ٢٠٢٤. وما إجمالي مدة التوقف بالساعات لكل شيفت؟"
--"What is the number of downtime events for each shift in 2024, along with the total downtime duration in hours for each shift?"
SELECT COUNT(Duration_hours) AS COUNT_OF_DOWNTIME, Shift ,  SUM(Duration_hours) AS TOTAL_DURATION_HOURS
FROM Machine_Downtime
GROUP BY  SHIFT
ORDER BY TOTAL_DURATION_HOURS DESC


--=========================================================================================================================


-- RELATION BETWEEN DURATION HOURS , SHIFT AND MONTH
--"ما عدد حالات التوقف لكل شيفت في كل شهر خلال عام ٢٠٢٤. وما إجمالي مدة التوقف بالساعات لكل مجموعة شهر-شيفت؟"
--"What is the number of downtime events for each shift in each month of 2024, along with the total downtime duration in hours for each month-shift c
SELECT COUNT(Duration_hours) AS COUNT_OF_DOWNTIME, Start_Down_month , Shift , SUM(Duration_hours) AS TOTAL_DURATION_HOURS
FROM Machine_Downtime
GROUP BY Start_Down_month , SHIFT
ORDER BY Start_Down_month


--=========================================================================================================================


--      RELATION BETWEEN DURATION HOURS AND COUNT OF DOWNTIME PER MACHINE NAME
--"ما إجمالي مدة التوقف بالساعات وعدد حالات التوقف لكل ماكينة خلال عام ٢٠٢٤؟"
--"What is the total downtime duration in hours and the number of downtime events for each machine in 2024?"

SELECT  Machine_Name , SUM(Duration_hours) AS TOTAL_DURATION_HOURS_PER_MACHINE , COUNT(Duration_hours) AS COUNT_OF_DOWNTIME
FROM Machine_Downtime
GROUP BY Machine_Name
ORDER BY TOTAL_DURATION_HOURS_PER_MACHINE DESC


--=========================================================================================================================
```

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Object Explorer

Connect

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- Security
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- XEvent Profiler

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```sql
--                                    MACHINE_INFORMATION SHEET


--      RELATIONSHIP BETWEEN MANUFACTURER AND TOTAL DURATION HOURS
--"ما المصنع الذي كانت ماكيناته الأكثر توقفًا من حيث إجمالي مدة التوقف في عام ٢٠٢٤؟"
--"Which manufacturer's machines experienced the most downtime in terms of total hours in 2024?"

SELECT  I.Manufacturer , SUM(D.Duration_hours) AS TOTAL_DURATION_HOURS
FROM Machine_Downtime D inner join Machine_Info I
ON D.Machine_ID = I.Machine_ID
GROUP BY  I.Manufacturer
ORDER BY TOTAL_DURATION_HOURS DESC


--===========================================================================================================



--"أي مصنع للماكينات يحتاج إلى أكبر قدر من الصيانة المخططة خلال عام ٢٠٢٤؟"
--"Which machine manufacturer required the most planned maintenance in 2024?"
SELECT  I.Manufacturer , SUM(D.Duration_hours) AS TOTAL_DURATION_HOURS
FROM Machine_Downtime D inner join Machine_Info I
ON D.Machine_ID = I.Machine_ID
where Planned = 'Maintenance'
GROUP BY  I.Manufacturer
ORDER BY TOTAL_DURATION_HOURS DESC


--===========================================================================================================



--"أي مصنع سجلت ماكيناته أكثر توقف غير مخطط بسبب الأعطال خلال عام ٢٠٢٤؟"
--"Which manufacturer had the most unplanned downtime due to failures in 2024?"
SELECT  I.Manufacturer , SUM(D.Duration_hours) AS TOTAL_DURATION_HOURS
FROM Machine_Downtime D inner join Machine_Info I
ON D.Machine_ID = I.Machine_ID
where Planned = 'Failure'
GROUP BY  I.Manufacturer
ORDER BY TOTAL_DURATION_HOURS DESC
```

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Ready Add to Source Control

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```sql
--     RELATIONSHIP BETWEEN MACHINE AGE , SUM OF DURATION HOURS AND MANTAINCE
--"ما إجمالي مدة التوقف المخطط وغير المخطط لكل ماكينة خلال عام ٢٠٢٤، مع ذكر عمر الماكينة والشركة المصنعة لها؟"
--"What is the total planned and unplanned downtime for each machine in 2024, including the machine's age and its manufacturer?"
SELECT
    D.Machine_Name,
    I.Machine_Age,
    I.Manufacturer,
    SUM(CASE WHEN D.Planned = 'Maintenance ' THEN D.Duration_hours ELSE 0 END) AS TOTAL_DURATION_HOURS_UNPLANNED,
    SUM(CASE WHEN D.Planned = 'Failure ' THEN D.Duration_hours ELSE 0 END) AS TOTAL_DURATION_HOURS_PLANNED
FROM Machine_Info I
INNER JOIN Machine_Downtime D
    ON I.Machine_ID = D.Machine_ID
GROUP BY
    D.Machine_Name,
    I.Machine_Age,
    I.Manufacturer

--=================================================================================================================

        RELATIONSHIP BETWEEN DURATION HOURS , TARGET RUNTIME HOURS DAY AND RUNTIME PER DAY
```

	Machine_Name	Machine_Age	Manufacturer	TOTAL_DURATION_HOURS_UNPLANNED	TOTAL_DURATION_HOURS_PLANNED
1	Stamping_Machine_031	9	Mitsubishi	792.866668701172	4331.6333770752
2	Assembly_Machine_014	1	KUKA	354.449999421835	588.999998182058
3	Painting_Machine_001	3	Fanuc	219.183332830667	773.683333903551
4	Stamping_Machine_024	1	ABB	183.016666591167	609.449999153614
5	Stamping_Machine_033	10	Schneider	797.666662216187	3652.1333770752
6	Assembly_Machine_027	3	Siemens	418.849999189377	672.066665768623
7	Stamping_Machine_020	1	ABB	165.533334732056	543.866667032242
8	Packaging_Machine_025	6	Siemens	343.883333534002	761.199999541044
9	Assembly_Machine_013	1	ABB	228.450001180172	582.850001484156
10	Stamping_Machine_035	1	KUKA	176.366666048765	777.266666382551
11	Stamping_Machine_015	4	Fanuc	211.883333027363	844.2999997437

Query executed successfully. | 30201091100258\SQLEXPRESS (... | 30201091100258\LENOVO ... | Manufactruing | 00:00:00 | 35 rows

Phase III: Deep Processing and Metric Creation (Python)

To transform the structured and aggregated data received from SQL into an analytical-ready format, and to compute crucial Key Performance Indicators (KPIs)

Key Implementation Steps

1 Library Import and Setup

- Imported essential libraries: Pandas for data manipulation, NumPy for numerical operations, and Matplotlib / plotly / Seaborn for visualization.

2 Data Acquisition

Read the final unified dataset (or merged the source datasets, if done in Python) into a Pandas DataFrame.

3 Exploratory Data Analysis (EDA)

- Structure Check: Used functions like shape, head, and tail to view the dimensions and initial records of the dataset.
- Column Information: Executed info() and describe() to understand data types, non-null counts, and key statistical summaries (mean, max, min, quartiles) for numeric columns.

4 Data Cleaning and Refinement

- Handling Missing Values: Verified and treated any potential Null values.
- Duplicate Removal: Used the drop_duplicates function to ensure data integrity by removing redundant records.
- Column Management: Dropped any unnecessary or redundant columns to streamline the dataset for analysis.

Python Execution (Part 2) — Core Analysis and Visualization

Objective: To execute the core analytical queries using powerful grouping and aggregation functions to generate the final KPIs and visual evidence for the presentation.

Key Implementation Steps

1 Basic Aggregation

- Calculated the fundamental metrics: Sum, Average, and Count for the Duration_hours and total downtime incidents.

2 Comparative Analysis (Grouping)

- Shift Analysis: Calculated total hours and count of incidents per Shift (A, B, C) and determined the Total Run Time per shift.
- Temporal Analysis: Grouped downtime by Month and Day to identify seasonal or weekly patterns.
- Department/Cause Analysis: Grouped by Downtime_Reason (to find top causes) and by Department (to find total hours lost).

3 Planned vs. Unplanned Analysis

- Created categories to distinguish between Planned Maintenance and Unplanned Failures.
- Calculated the total count and hours of Planned vs. Unplanned downtime per machine.

4 Operator and Machine Performance

- Calculated the total stoppage count per machine relative to the responsible Operator.
- Created a composite metric showing Operator performance broken down by Planned vs. Unplanned stops.

5 Final Visualization

- Generated all core charts, including:
- Bar Charts showing downtime hours by cause and efficiency scores.
- Pie/Bar Charts comparing Planned vs. Unplanned stops per machine/department.

The Final Phase: Building the Interactive Dashboard with Power BI

Objective: The ultimate goal of using Power BI was to translate the complex statistical results generated by Python, and the final charts into an interactive and user-friendly tool. This allows decision-makers to visually explore the data and apply various time and operational filters seamlessly.

Key Implementation Steps

1 Final Data Import

- The clean, engineered dataset, which now includes the new performance indicators (Efficiency %, Machine Age), was imported directly from the SQL database or the processed Python output.

2 Data Modeling and Relationships

- We ensured the establishment of correct data relationships within the Power BI model. This step is critical to guarantee that filters (Slicers) applied by the user correctly cascade and affect all visual elements on the dashboard.

3 Visual Design and KPI Presentation

- Dashboard was divided into four main Parts:

1- Holistic View of all Manfucteig Downtime Metrics

2-Detailed Shift -Based Analysis of Runtime and Downtime Patterns

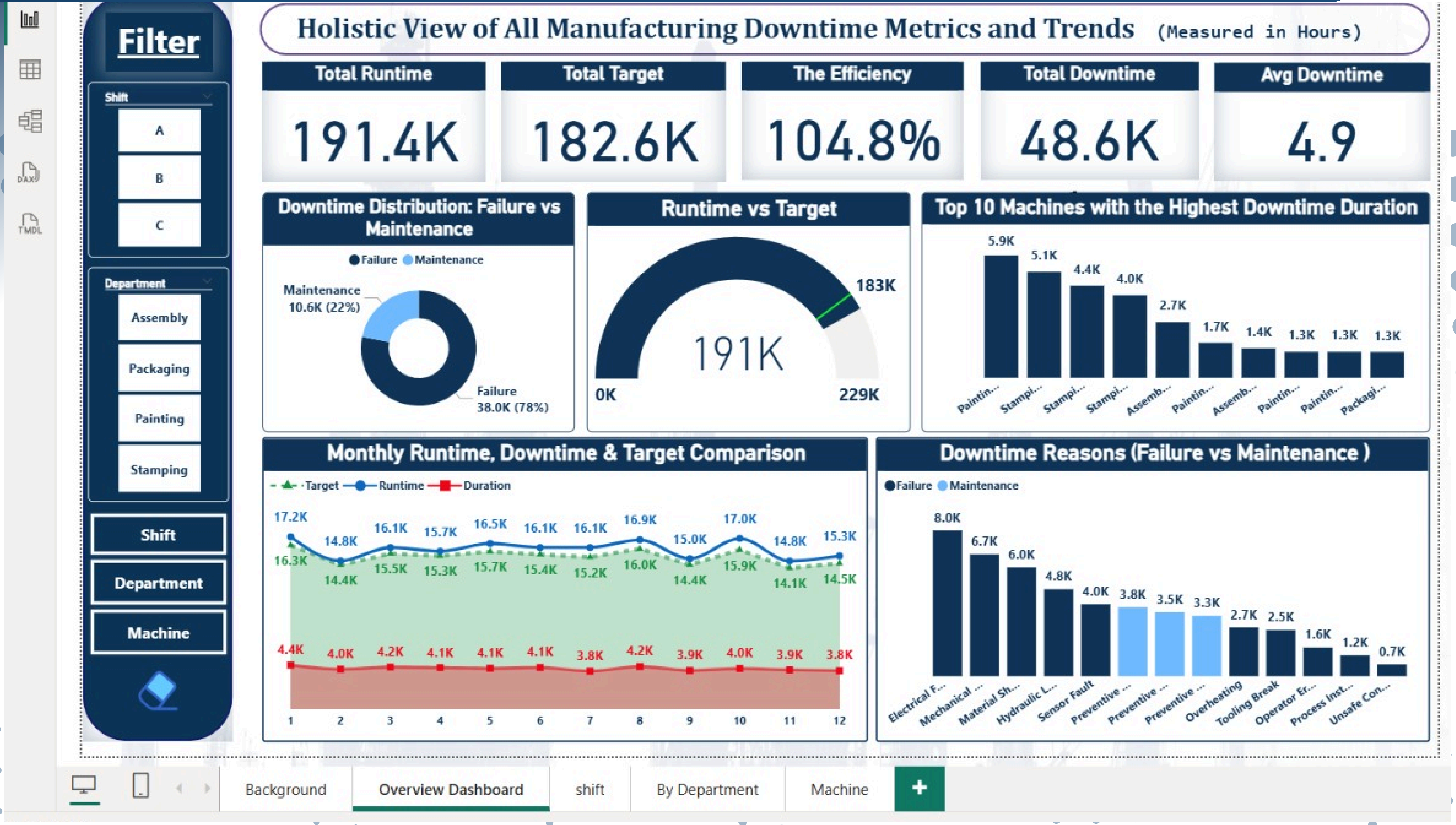
3-Department -Level Breakdown of Downtime Causes, Duration, and Frequency

4-Machine - Specific Performance Analysis of Runtime, Downtime and Failure Trands

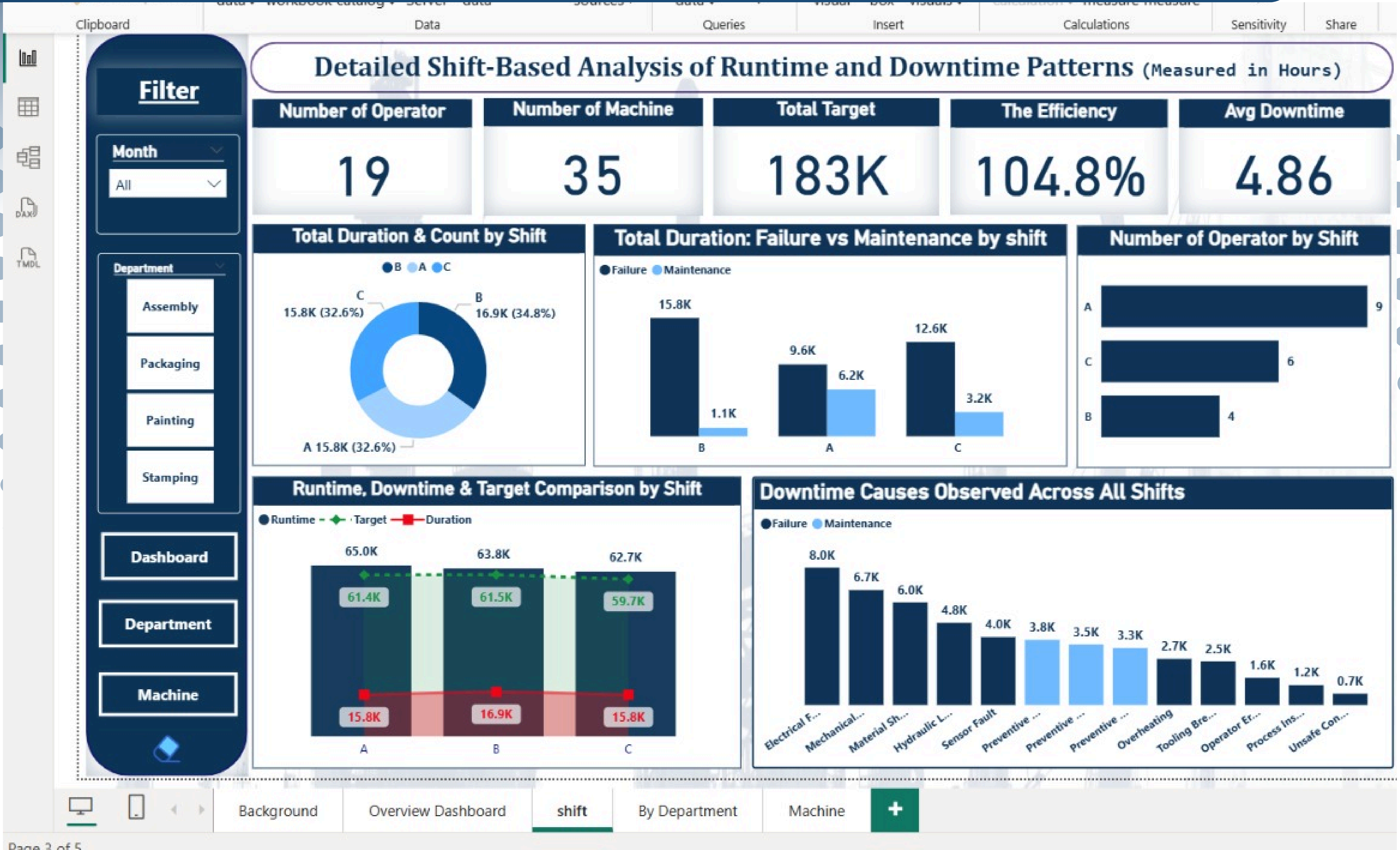
4 Enabling Interactivity

Implemented slicers and filters allowing users to dynamically segment the results.

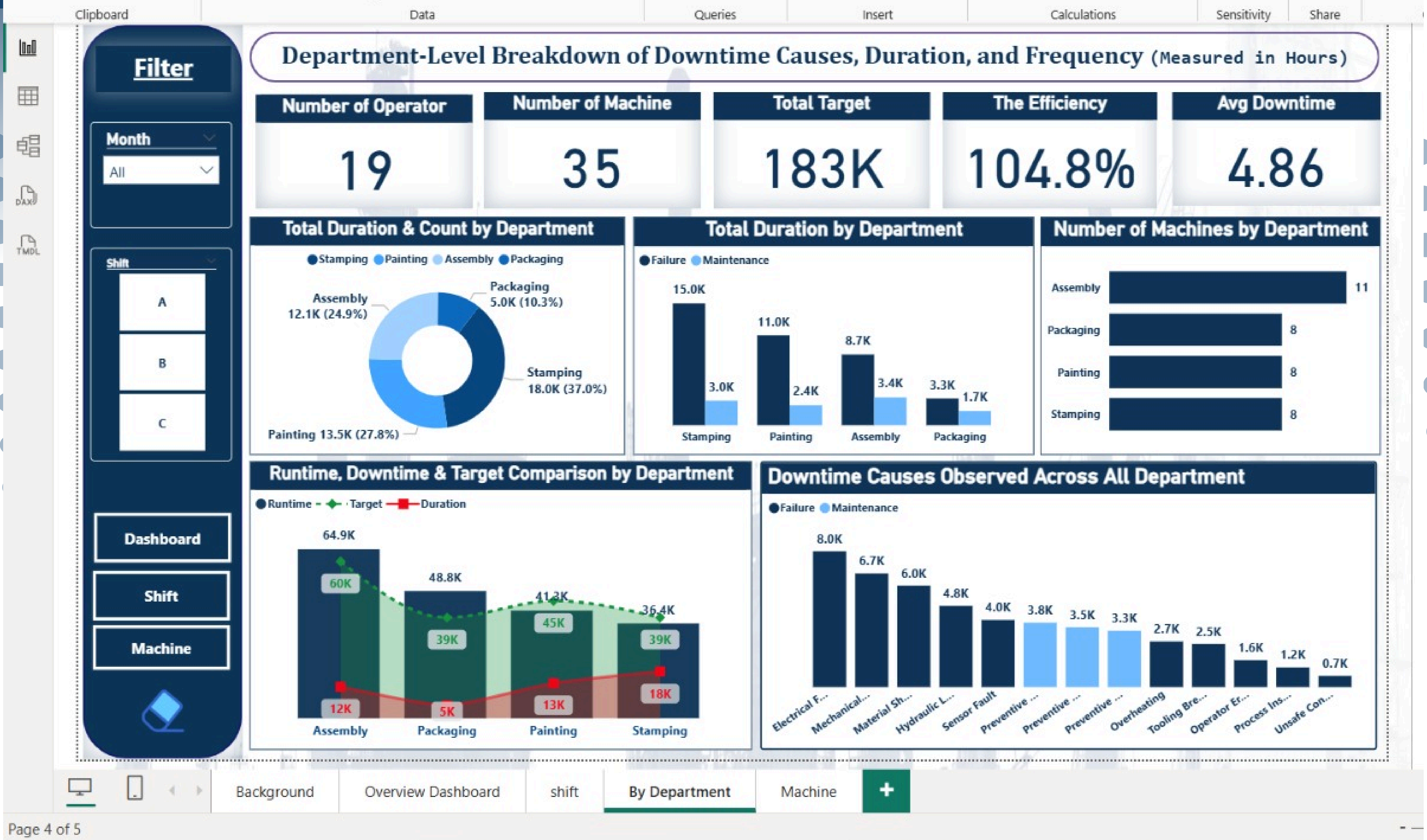
Holistic View of all Manfucteig Downtime Metrics



Detailed Shift -Based Analysis of Runtime and Downtime Patterns



Department -Level Breakdown of Downtime Causes, Duration, and Frequency



Machine - Specific Performance Analysis of Runtime, Downtime and Failure Trands

Power BI

Machine-Specific Performance Analysis of Runtime, Downtime, and Failure Trends (Measured in Hours)

Filter

Month: All

Department: All

Shift: All

Manufacturer: All

Machine_Name	Manufacturer	Machine_Age	Maint_Count_Y_hr	Maint_Year_Hrs	Maint_Average_Y:hrs	Failure_Count_Y	Failures_year	Failure_Average_Y	Runtime_Status	Efficiency
Painting_Machine_021	Schneider	13	50	697	13.93	265	5220	19.70	Poor	23.71%
Stamping_Machine_029	Schneider	9	53	436	8.22	232	3560	15.35	Poor	45.36%
Stamping_Machine_031	Mitsubishi	9	63	793	12.59	267	4332	16.22	Poor	47.06%
Stamping_Machine_033	Schneider	10	59	798	13.52	227	3652	16.09	Poor	46.90%
Assembly_Machine_007	Mitsubishi	9	74	395	5.33	228	2297	10.08	Good	68.57%
Assembly_Machine_010	Mitsubishi	6	95	451	4.75	216	982	4.55	Good	88.15%
Painting_Machine_016	Siemens	6	68	296	4.35	216	962	4.45	Good	88.97%
Painting_Machine_022	Mitsubishi	6	72	332	4.62	192	926	4.82	Good	87.43%
Painting_Machine_028	Siemens	6	74	321	4.34	192	818	4.26	Good	89.63%
Assembly_Machine_027	Siemens	3	98	419	4.27	169	672	3.98	Excellent	99.57%
Assembly_Machine_032	Siemens	6	98	412	4.20	196	825	4.21	Excellent	98.96%
Packaging_Machine_017	Siemens	7	104	431	4.14	196	820	4.18	Excellent	99.16%
Packaging_Machine_025	Siemens	6	85	344	4.05	179	761	4.25	Excellent	99.07%
Painting_Machine_005	Siemens	6	79	270	3.42	220	1427	6.48	Excellent	91.62%
Assembly_Machine_011	ABB	1	82	221	2.69	193	564	2.92	Amazing	151.05%
Assembly_Machine_012	Fanuc	3	109	136	1.25	190	458	2.41	Amazing	110.06%
Assembly_Machine_013	ABB	1	81	228	2.82	205	583	2.84	Amazing	151.17%
Assembly_Machine_014	KUKA	1	107	354	3.31	191	589	3.08	Amazing	130.21%
Assembly_Machine_019	Fanuc	3	101	93	0.92	192	330	1.72	Amazing	102.52%
Assembly_Machine_023	Fanuc	3	95	345	3.63	209	695	3.33	Amazing	128.62%
Assembly_Machine_026	Fanuc	3	93	364	3.91	186	693	3.73	Amazing	101.05%
Packaging_Machine_002	KUKA	2	101	23	0.23	176	81	0.46	Amazing	131.25%
Packaging_Machine_003	ABB	1	90	266	2.95	180	509	2.83	Amazing	150.92%
Packaging_Machine_004	KUKA	1	104	20	0.20	182	44	0.24	Amazing	148.59%
Packaging_Machine_006	Fanuc	3	85	9	0.10	188	32	0.17	Amazing	108.41%
Packaging_Machine_018	ABB	1	108	308	2.85	181	525	2.90	Amazing	150.86%
Packaging_Machine_034	ABB	1	107	330	3.08	178	504	2.83	Amazing	150.52%
Painting_Machine_001	Fanuc	3	69	219	3.18	204	774	3.79	Amazing	113.13%
Painting_Machine_008	KUKA	1	76	244	3.22	229	748	3.27	Amazing	129.67%
Painting_Machine_009	KUKA	1	105	70	0.66	173	173	1.00	Amazing	144.55%
Stamping_Machine_015	Fanuc	4	57	212	3.72	223	844	3.79	Amazing	112.38%

Dashboard

Shift

Department

Background | Overview Dashboard | shift | By Department | Machine | +

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Insight Makers
MANUFACTURING DOWNTIME

Thank you